

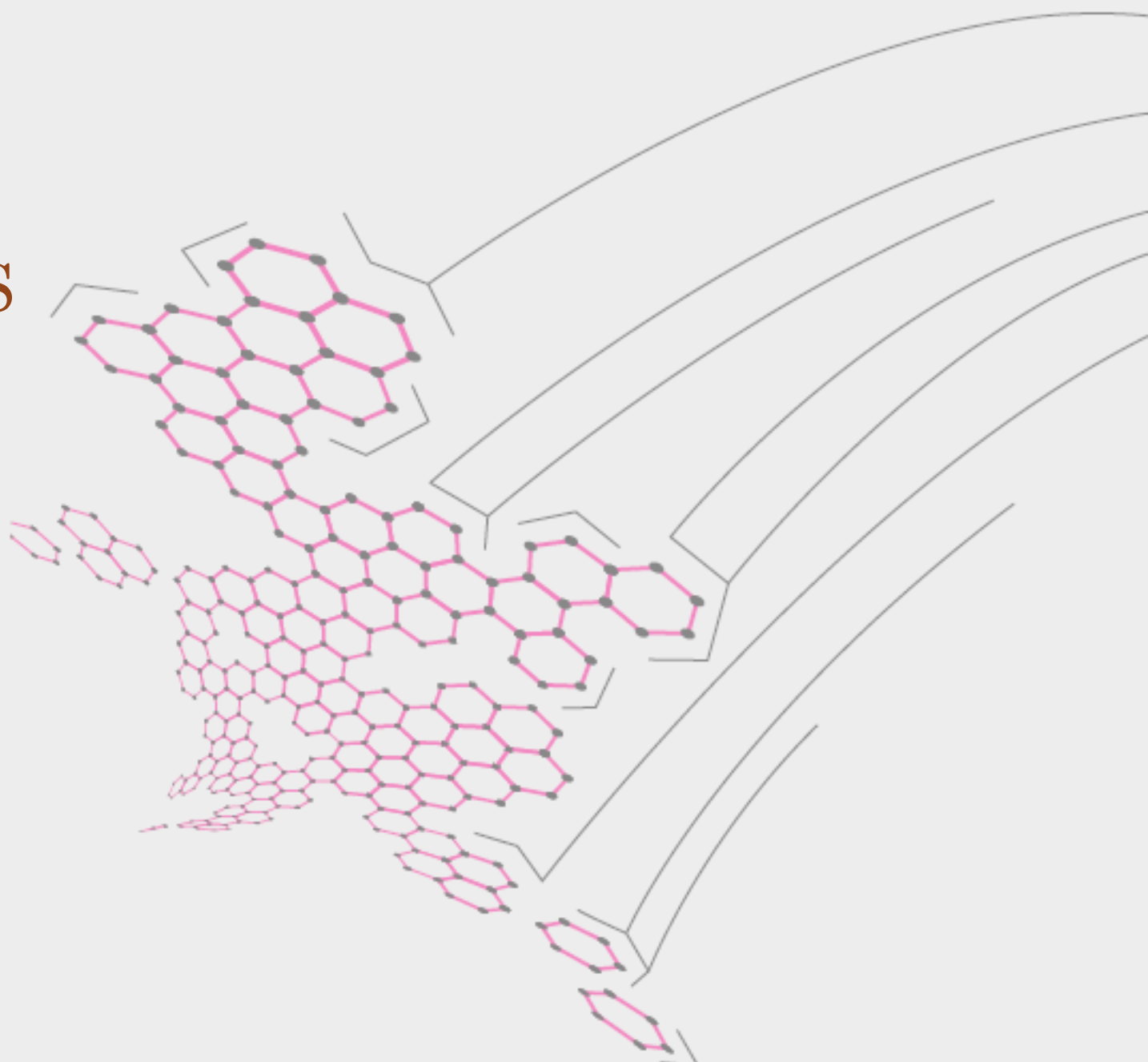
Calculus A: Lecture 3

Integration of single-variable functions

Carlos Pérez Arancibia
(c.a.perezarancibia@utwente.nl)

Mathematics of Computational Science
Department of Applied Mathematics
University of Twente

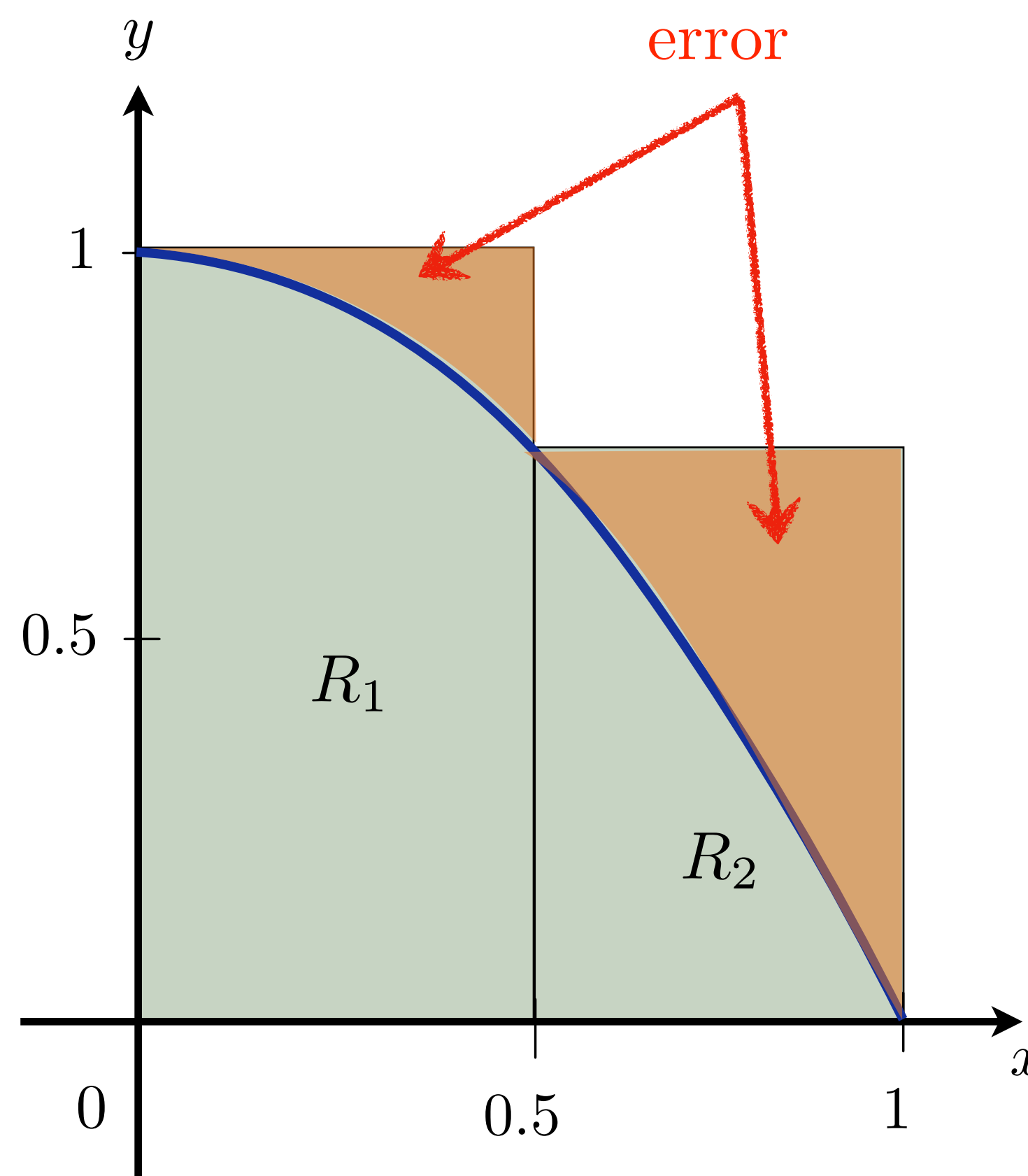
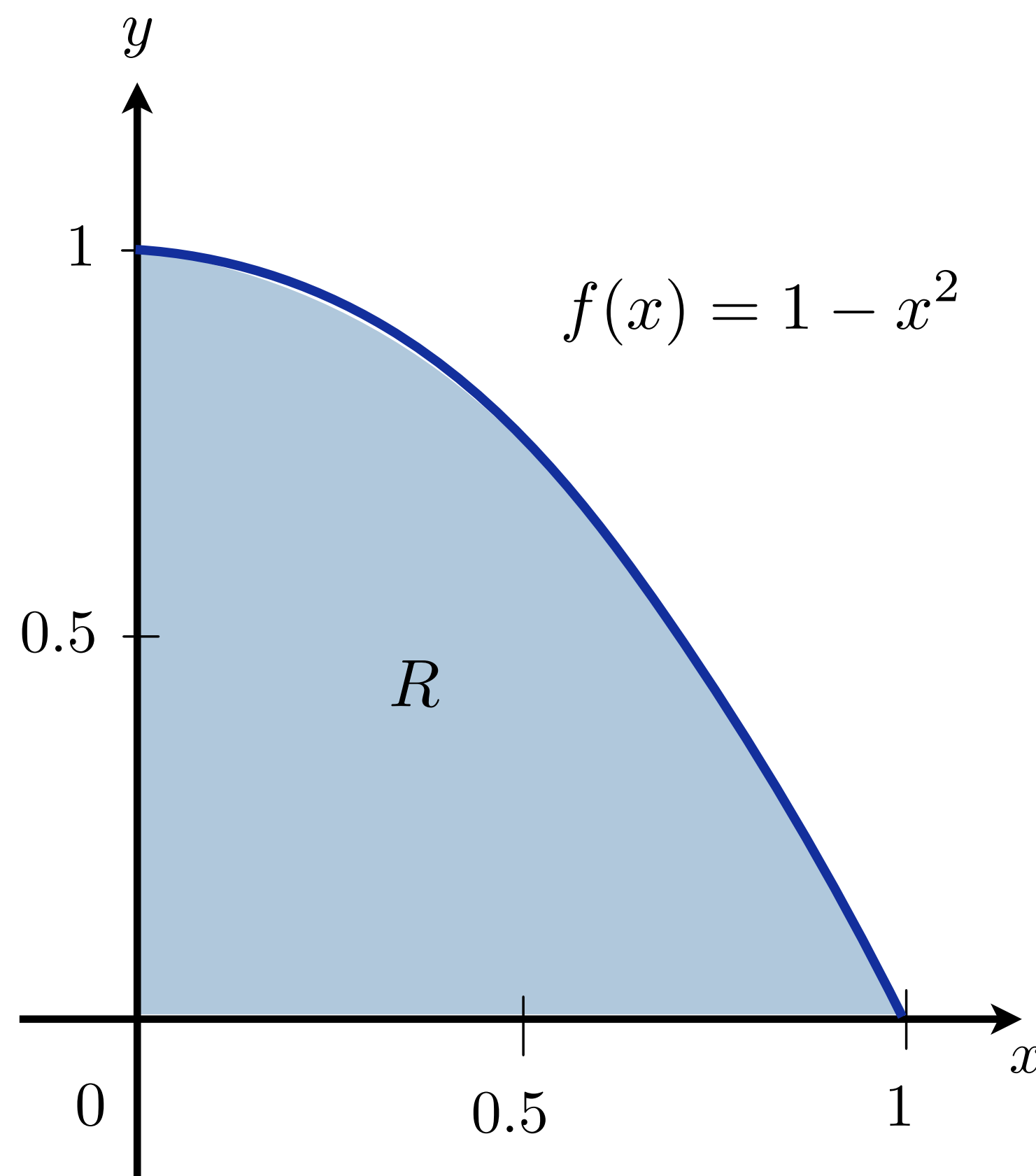
September 14th, 2026



Integration

The **definite integral** rigorously defines quantities like **area** and **volume** by **summing** approximations and taking their **limit**, with broad applications in mathematics and science.

How can we calculate the **area** of R ?



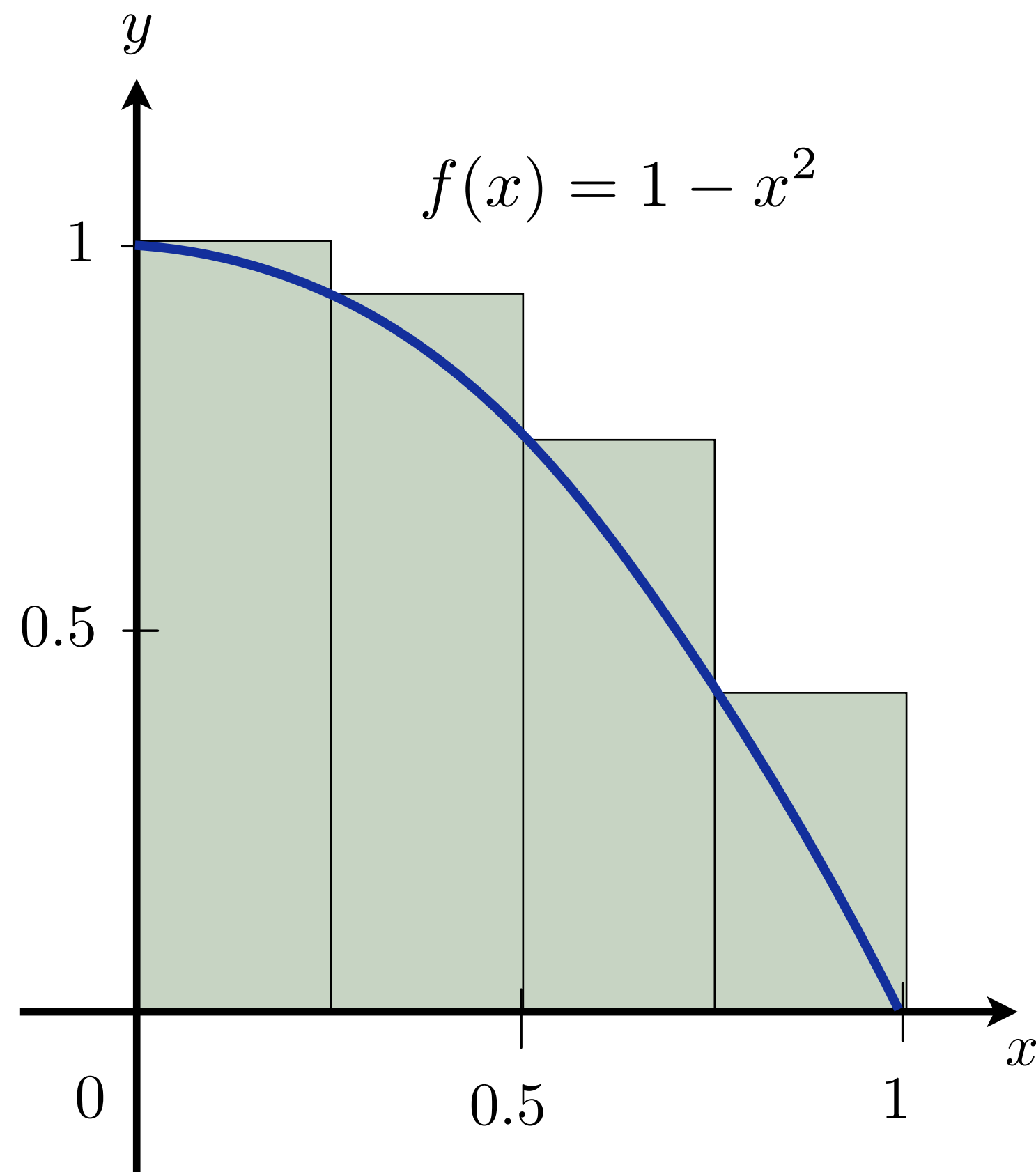
$$\begin{aligned} A(R) &\approx \underbrace{\frac{1}{2}f(0)}_{A(R_1)} + \underbrace{\frac{1}{2}f(\frac{1}{2})}_{A(R_2)} \\ &= \frac{1}{2} \cdot 1 + \frac{1}{2} \cdot \left(1 - \frac{1}{4}\right) \\ &= \frac{7}{8} \end{aligned}$$

How do we make the error smaller?

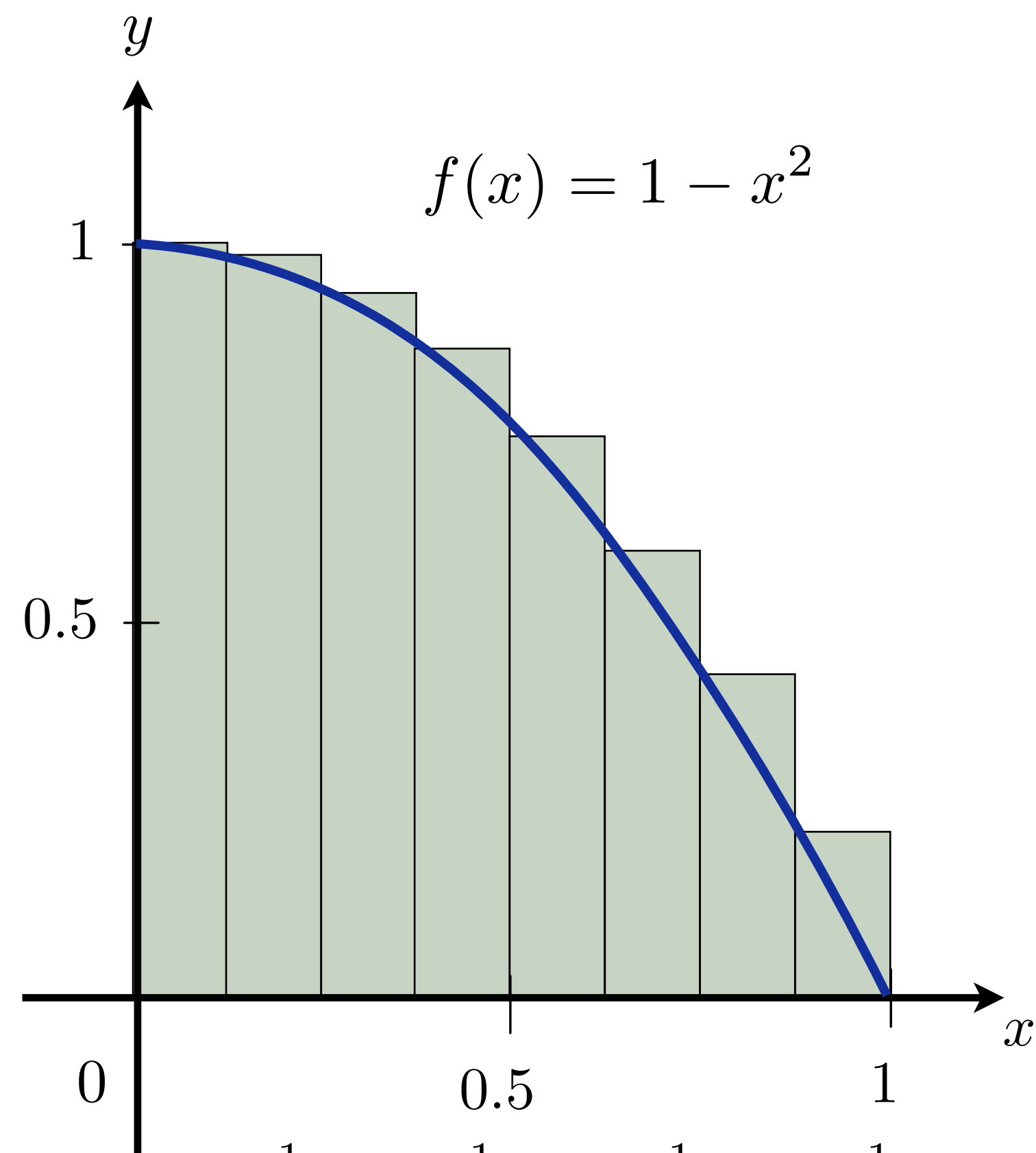
Integration

The **definite integral** rigorously defines quantities like **area** and **volume** by **summing** approximations and taking their **limit**, with broad applications in mathematics and science.

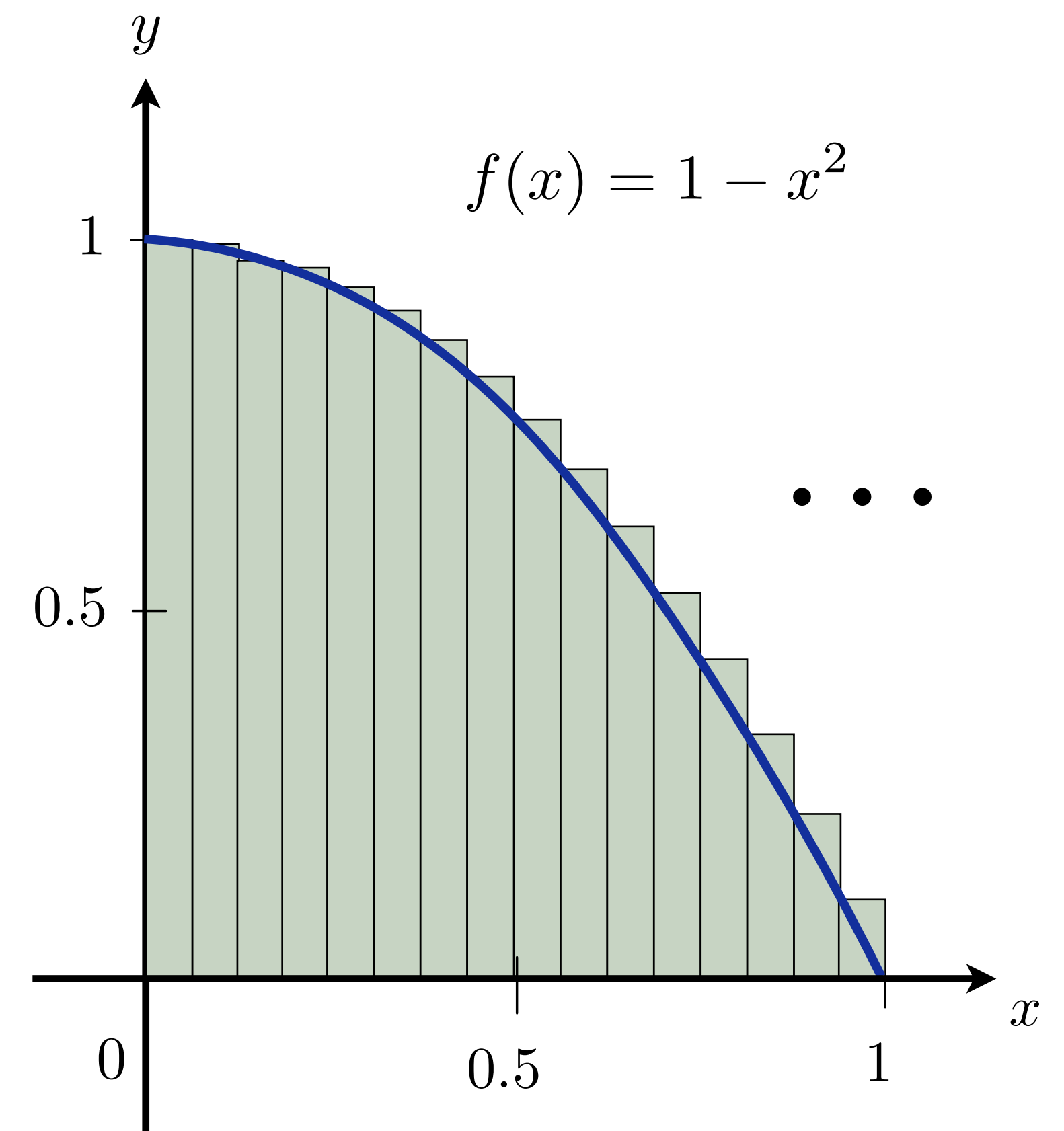
How can we calculate the **area** of R ?



$$A(R) \approx \frac{1}{4}f(0) + \frac{1}{4}f\left(\frac{1}{4}\right) + \frac{1}{4}f\left(\frac{1}{2}\right) + \frac{1}{4}f\left(\frac{3}{4}\right)$$



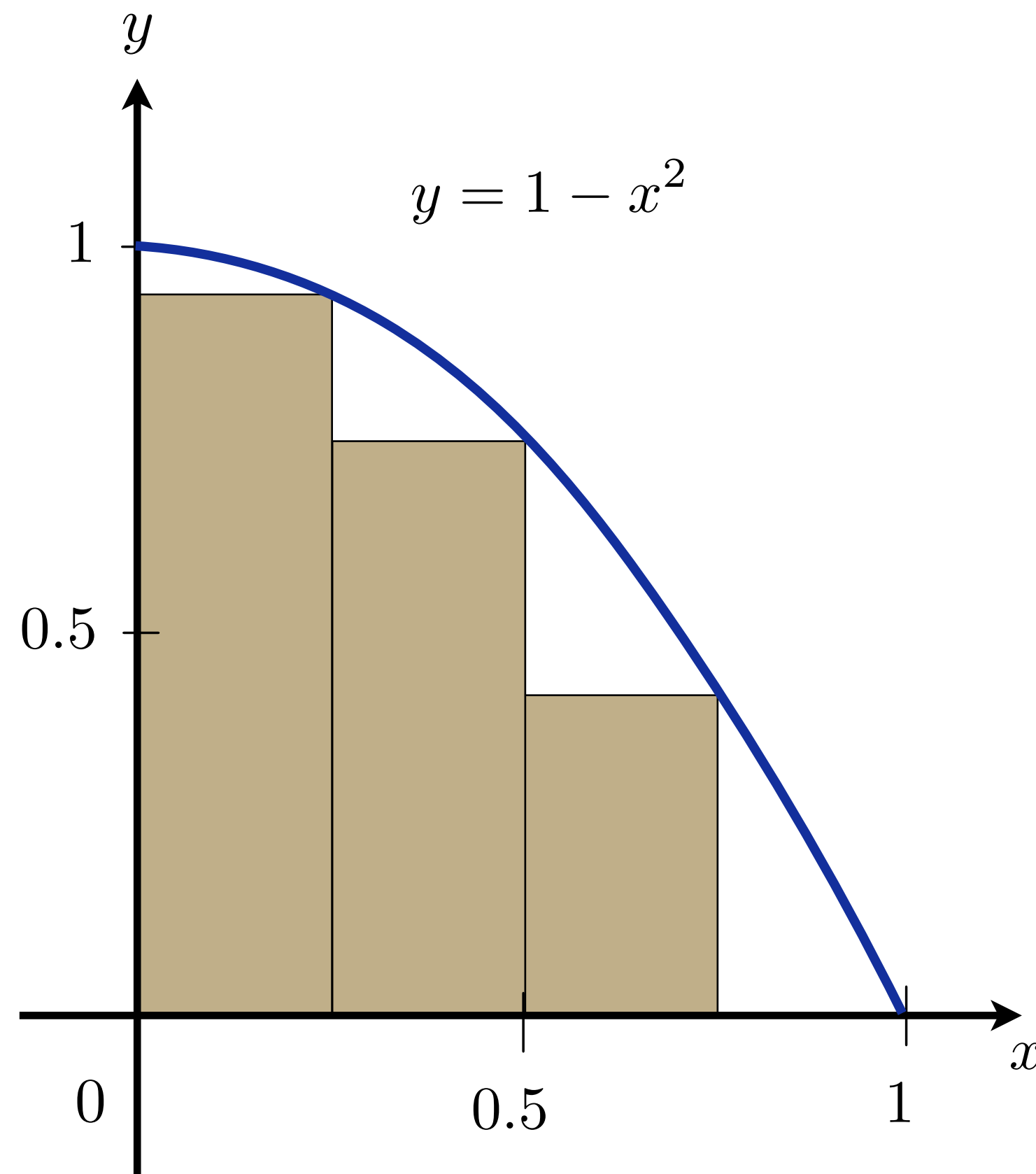
$$A(R) \approx \frac{1}{8}f(0) + \frac{1}{8}f\left(\frac{1}{8}\right) + \frac{1}{8}f\left(\frac{1}{4}\right) + \frac{1}{8}f\left(\frac{3}{8}\right) + \\ \frac{1}{8}f\left(\frac{1}{2}\right) + \frac{1}{8}f\left(\frac{5}{8}\right) + \frac{1}{8}f\left(\frac{3}{4}\right) + \frac{1}{8}f\left(\frac{7}{8}\right)$$



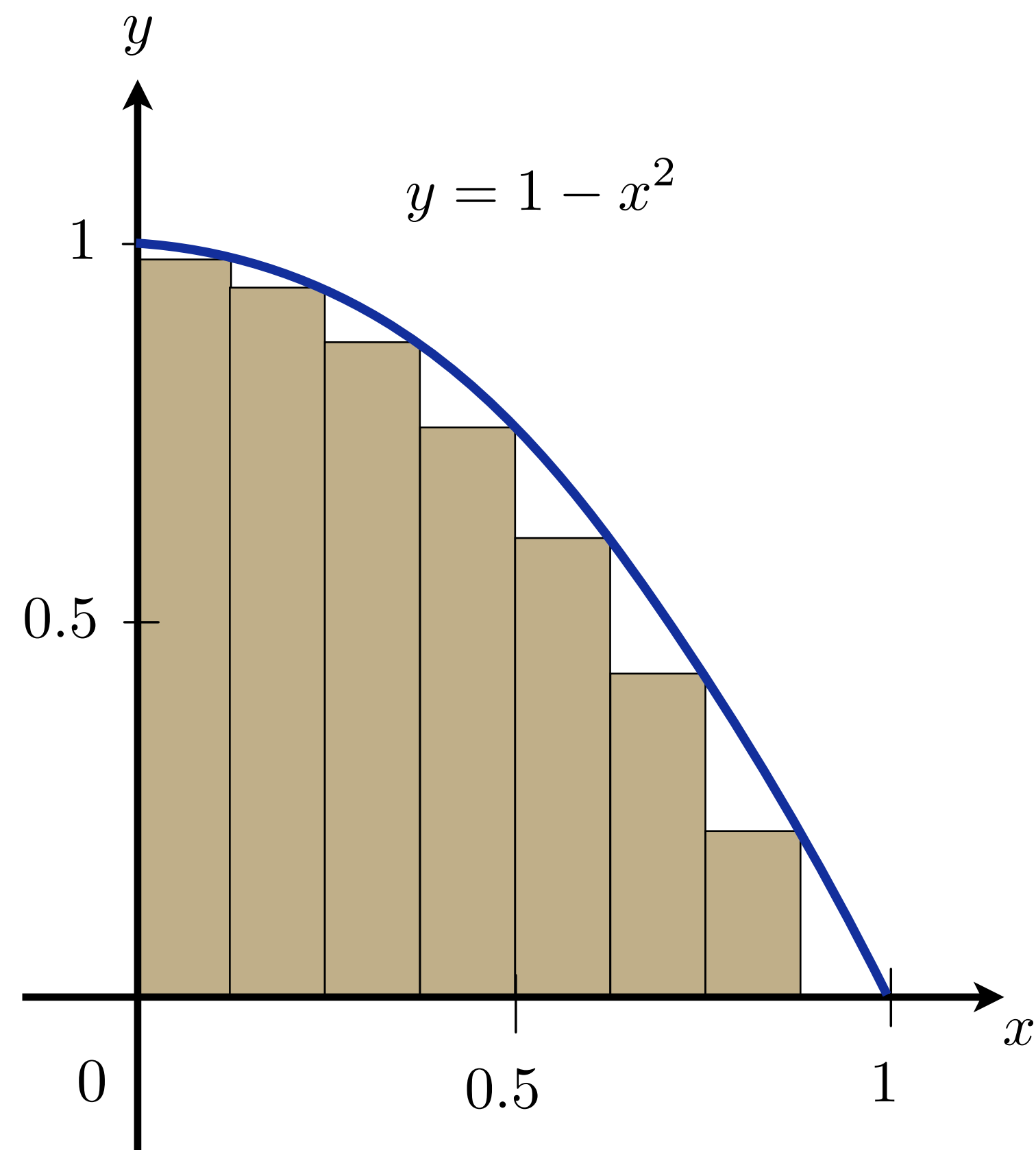
Integration

The **definite integral** rigorously defines quantities like **area** and **volume** by **summing** approximations and taking their **limit**, with broad applications in mathematics and science.

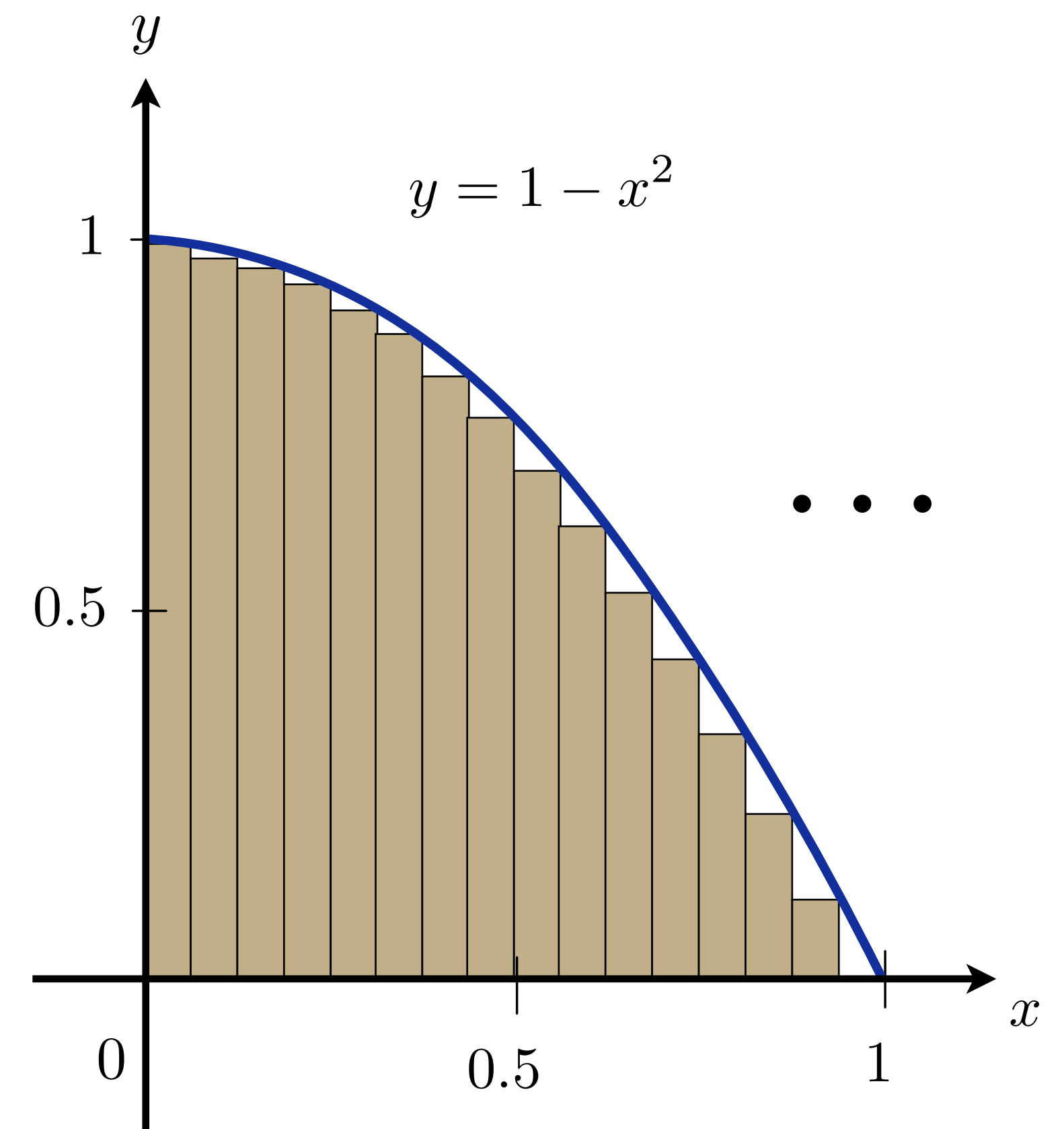
How can we calculate the **area** of R ?



$$A(R) \approx \frac{1}{4}f\left(\frac{1}{4}\right) + \frac{1}{4}f\left(\frac{1}{2}\right) + \frac{1}{4}f\left(\frac{3}{4}\right) + \frac{1}{4}f(1)$$



$$A(R) \approx \frac{1}{8}f\left(\frac{1}{8}\right) + \frac{1}{8}f\left(\frac{1}{4}\right) + \frac{1}{8}f\left(\frac{3}{8}\right) + \frac{1}{8}f\left(\frac{1}{2}\right) + \frac{1}{8}f\left(\frac{5}{8}\right) + \frac{1}{8}f\left(\frac{3}{4}\right) + \frac{1}{8}f\left(\frac{7}{8}\right) + \frac{1}{8}f(1)$$



...

Sigma Notation and Limits of Finite Sums

Definition: Sigma notation provides a compact way to represent a sum of terms.

Components:

$$\sum_{k=m}^n a_k = a_m + a_{m+1} + \cdots + a_n$$

k : index of summation
 n : ending value
 m : starting value
 a_k : general term

Example:

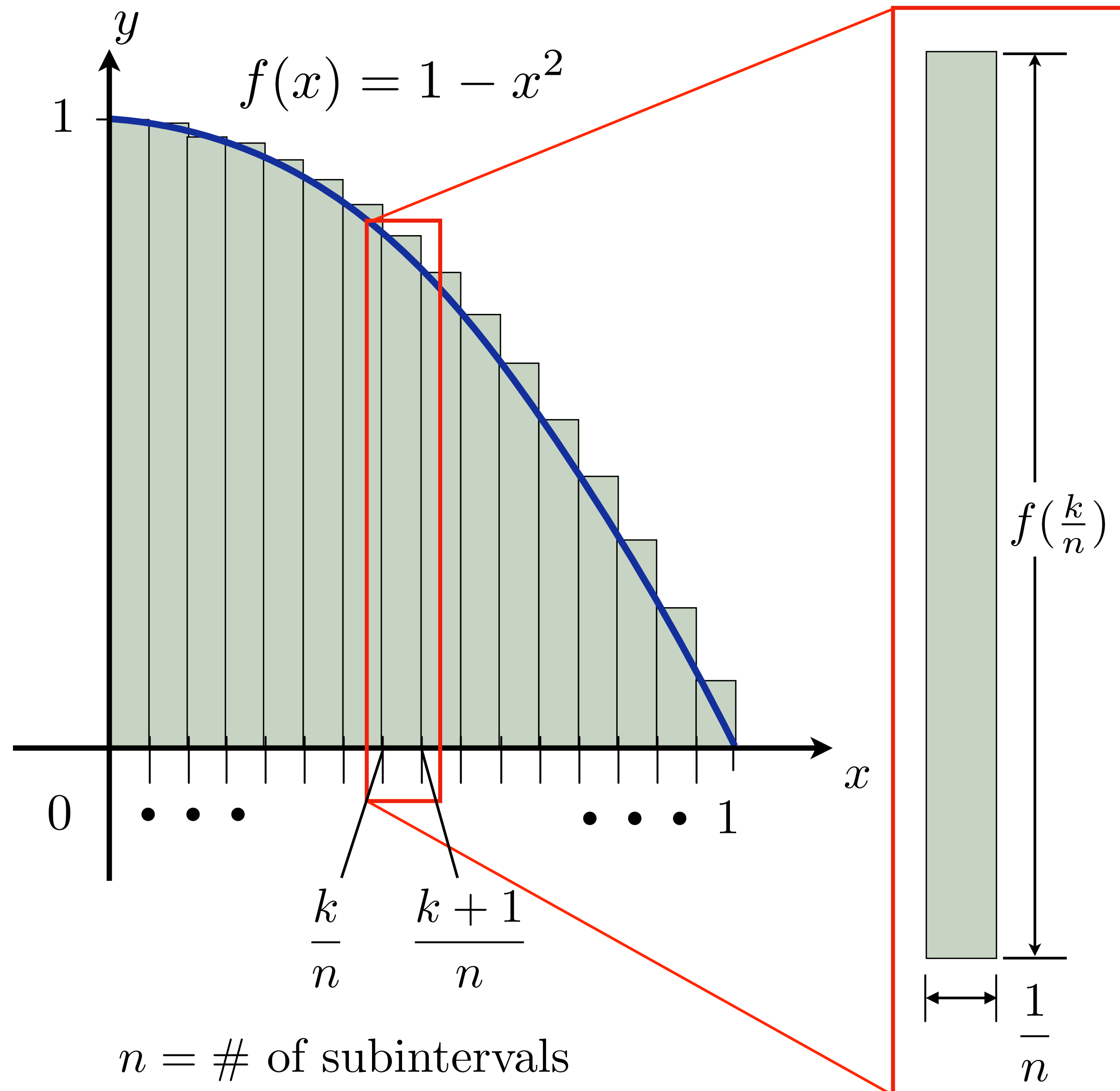
$$\sum_{k=1}^4 k^2 = 1^2 + 2^2 + 3^2 + 4^2 = 30$$

In general, the sums of the first n integers, squares, and cubes satisfy:

$$\sum_{k=1}^n k = \frac{n(n+1)}{2}, \quad \sum_{k=1}^n k^2 = \frac{n(n+1)(2n+1)}{6}, \quad \sum_{k=1}^n k^3 = \left[\frac{n(n+1)}{2} \right]^2.$$

Integration

To compute the exact area, we take the **limit** as the number of subintervals approaches infinity.



$$\begin{aligned}
 A(R) &= \lim_{n \rightarrow \infty} \sum_{k=1}^n \frac{1}{n} f\left(\frac{k}{n}\right) = \lim_{n \rightarrow \infty} \sum_{k=1}^n \frac{1}{n} \left(1 - \left(\frac{k}{n}\right)^2\right) \\
 &= \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{k=1}^n \left(1 - \frac{k^2}{n^2}\right) \\
 &= \lim_{n \rightarrow \infty} \frac{1}{n} \left(\sum_{k=1}^n 1 - \frac{1}{n^2} \sum_{k=1}^n k^2 \right) \\
 &= \lim_{n \rightarrow \infty} \frac{1}{n} \left(n - \frac{1}{n^2} \cdot \frac{n(n+1)(2n+1)}{6} \right) \\
 &= \frac{2}{3}
 \end{aligned}$$

The Definite Integral

Let $[a, b]$ be a closed interval on the real line. A **partition** P of $[a, b]$ is a finite ordered set of points

$$P = \{x_0, x_1, x_2, \dots, x_n\}$$

such that

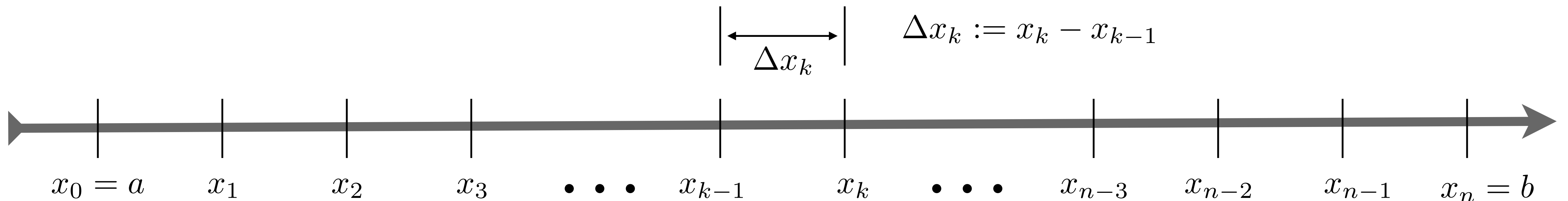
$$a = x_0 < x_1 < x_2 < \dots < x_n = b.$$

These points split $[a, b]$ into the **subintervals**

$$[x_0, x_1], [x_1, x_2], \dots, [x_{n-1}, x_n].$$

The **mesh size** of the partition P , denoted by $\|P\|$, is the length of the largest subinterval:

$$\|P\| = \max_{1 \leq k \leq n} (x_k - x_{k-1}) = \max_{1 \leq k \leq n} \Delta x_k.$$



The Definite Integral

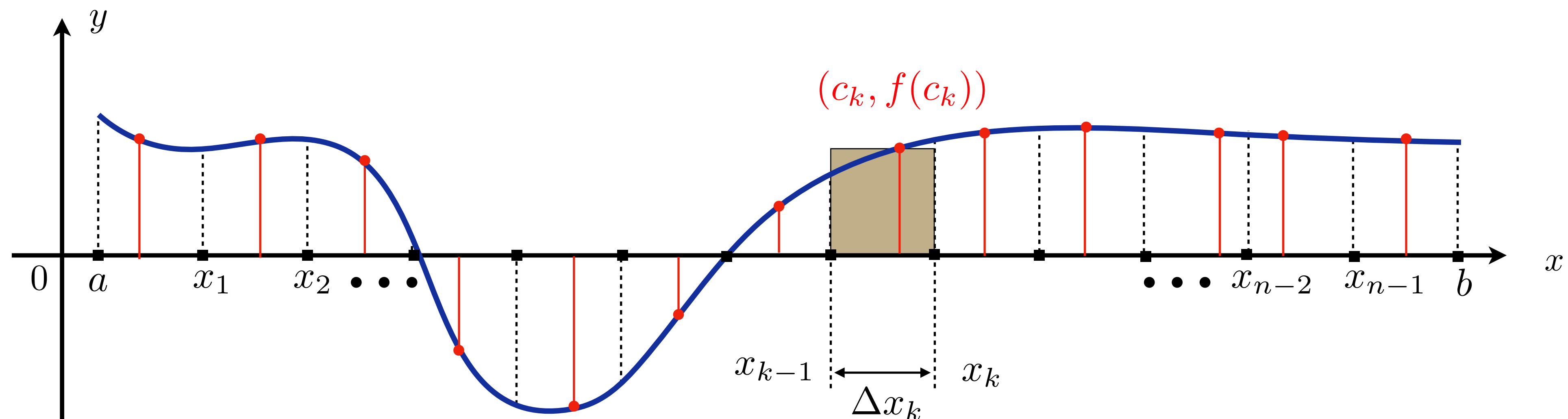
If the sums of the form

$$\sum_{k=1}^n f(c_k) \Delta x_k$$

over any **partition** $P = \{x_0, x_1, \dots, x_n\}$ of $[a, b]$ converge to the same limit I as the **mesh size** $\|P\|$ goes to zero, we say that f is **integrable** on $[a, b]$ and define

Definite Integral: $\lim_{\|P\| \rightarrow 0} \sum_{k=1}^n f(c_k) \Delta x_k = I = \int_a^b f(x) dx,$

where $\Delta x_k = x_k - x_{k-1}$ and $c_k \in [x_{k-1}, x_k]$ is an arbitrary sample point in the k -th subinterval.



The Definite Integral

upper limit of integration

integral sign
(elongated letter “S”)

lower limit of integration

function (integrand)

variable of integration

integral of f from a to b

The diagram illustrates the components of the definite integral notation $\int_a^b f(x) dx$. The integral sign is a large, blue, elongated letter 'S'. The upper limit b is at the top right of the sign, and the lower limit a is at the bottom right. The function $f(x)$ is the integrand, and dx is the variable of integration. A blue bracket underneath the entire expression $\int_a^b f(x) dx$ indicates that it represents the integral of f from a to b . Arrows point from the text labels to the corresponding parts of the notation.

The Definite Integral

Rules satisfied by definite integrals: For all f and g are integrable over $[a, b]$:

1. **Order of Integration:** $\int_b^a f(x) dx = - \int_a^b f(x) dx$ (Definition)

2. **Zero Width Interval:** $\int_a^a f(x) dx = 0$ (Definition, when $f(a)$ exists)

3. **Constant Multiple:** $\int_a^b k f(x) dx = k \int_a^b f(x) dx, \quad k \in \mathbb{R}$

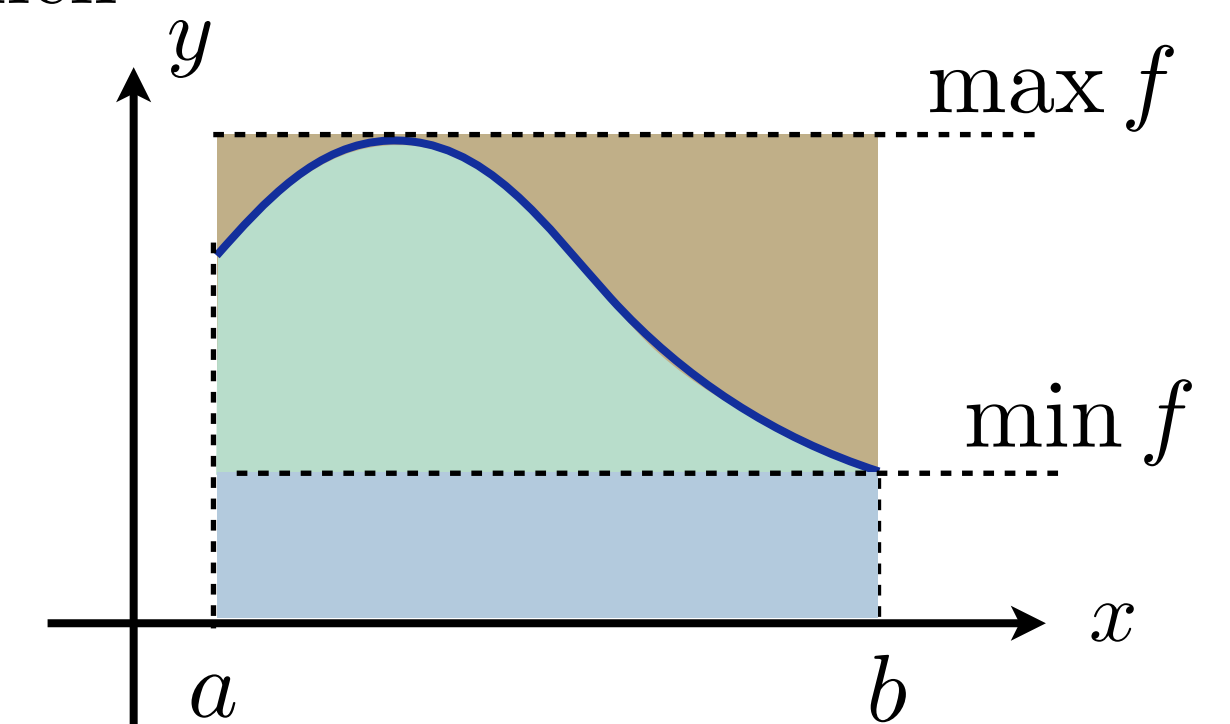
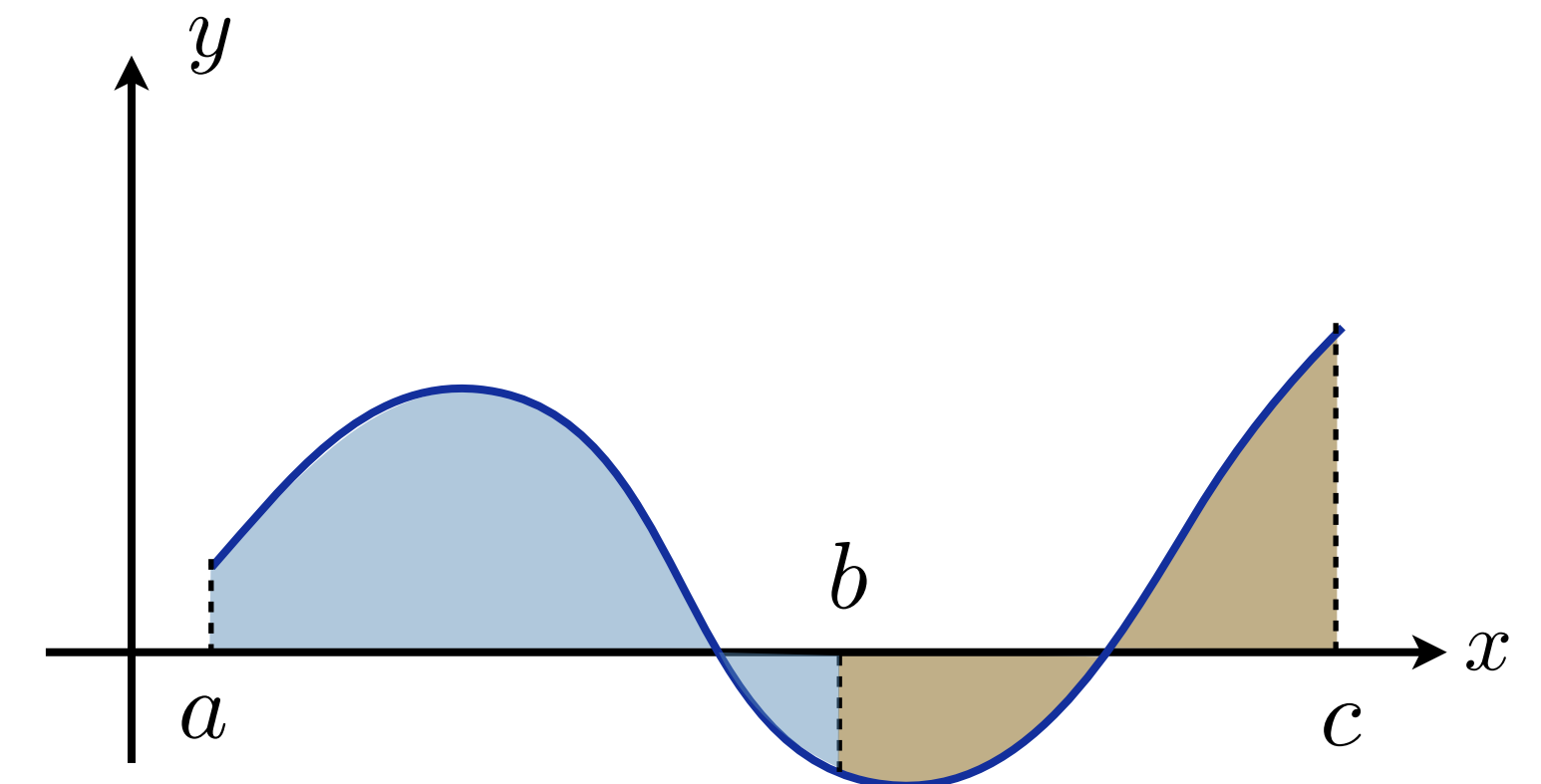
4. **Sum and Difference:** $\int_a^b (f(x) \pm g(x)) dx = \int_a^b f(x) dx \pm \int_a^b g(x) dx$

5. **Additivity:** $\int_a^b f(x) dx + \int_b^c f(x) dx = \int_a^c f(x) dx$

6. **Max–Min Inequality:** If f has maximum $\max f$ and minimum $\min f$ on $[a, b]$, then

$$\min f \cdot (b - a) \leq \int_a^b f(x) dx \leq \max f \cdot (b - a)$$

7. **Domination:** $f(x) \geq g(x)$ on $[a, b] \implies \int_a^b f(x) dx \geq \int_a^b g(x) dx.$



The Definite Integral

Example: Suppose that

$$\int_{-1}^1 f(x) \, dx = 5, \quad \int_1^4 f(x) \, dx = -2, \quad \text{and} \quad \int_{-1}^1 h(x) \, dx = 7.$$

Then:

1.

$$\int_4^1 f(x) \, dx = - \int_1^4 f(x) \, dx = -(-2) = 2 \quad (\text{Rule 1})$$

2.

$$\begin{aligned} \int_{-1}^1 [2f(x) + 3h(x)] \, dx &= 2 \int_{-1}^1 f(x) \, dx + 3 \int_{-1}^1 h(x) \, dx \\ &= 2(5) + 3(7) = 31 \quad (\text{Rules 3 and 4}) \end{aligned}$$

3.

$$\int_{-1}^4 f(x) \, dx = \int_{-1}^1 f(x) \, dx + \int_1^4 f(x) \, dx = 5 + (-2) = 3 \quad (\text{Rule 5})$$

The Definite Integral

Definition: If $y = f(x)$ is **nonnegative** and **integrable** over a closed interval $[a, b]$, then the **area under the curve $y = f(x)$ over $[a, b]$** is the integral of f from a to b ,

$$A = \int_a^b f(x) dx$$

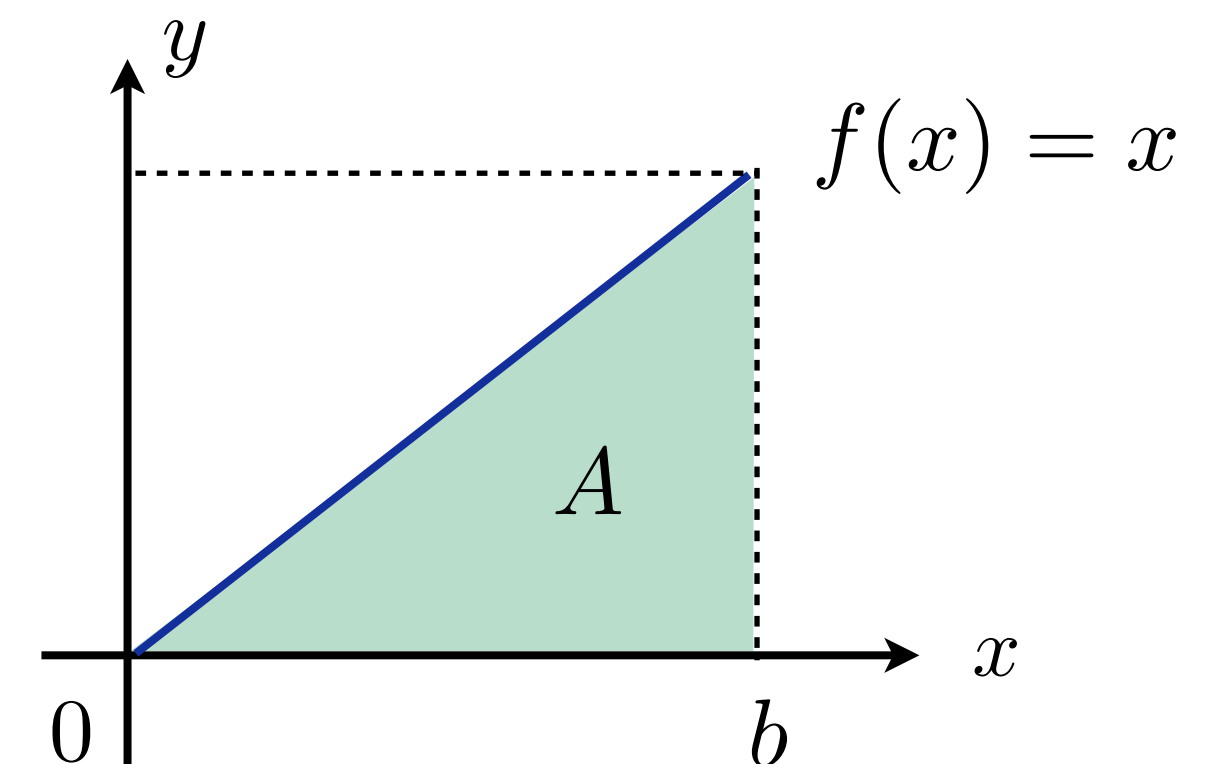
Example: Compute $\int_0^b x dx$ and find the area A under $y = x$ over the interval $[0, b]$, $b > 0$.

Solution. Partition $[0, b]$ into n subintervals of width $\Delta x = \frac{b}{n}$, and choose right endpoints $c_k = k\Delta x$:

$$S_n = \sum_{k=1}^n f(c_k) \Delta x = \sum_{k=1}^n (k\Delta x) \Delta x = (\Delta x)^2 \sum_{k=1}^n k = (\Delta x)^2 \frac{n(n+1)}{2}.$$

Substitute $\Delta x = b/n$ and take the limit as $n \rightarrow \infty$:

$$\int_0^b x dx = \lim_{n \rightarrow \infty} S_n = \lim_{n \rightarrow \infty} \frac{b^2(n+1)}{2n} = \frac{b^2}{2}.$$



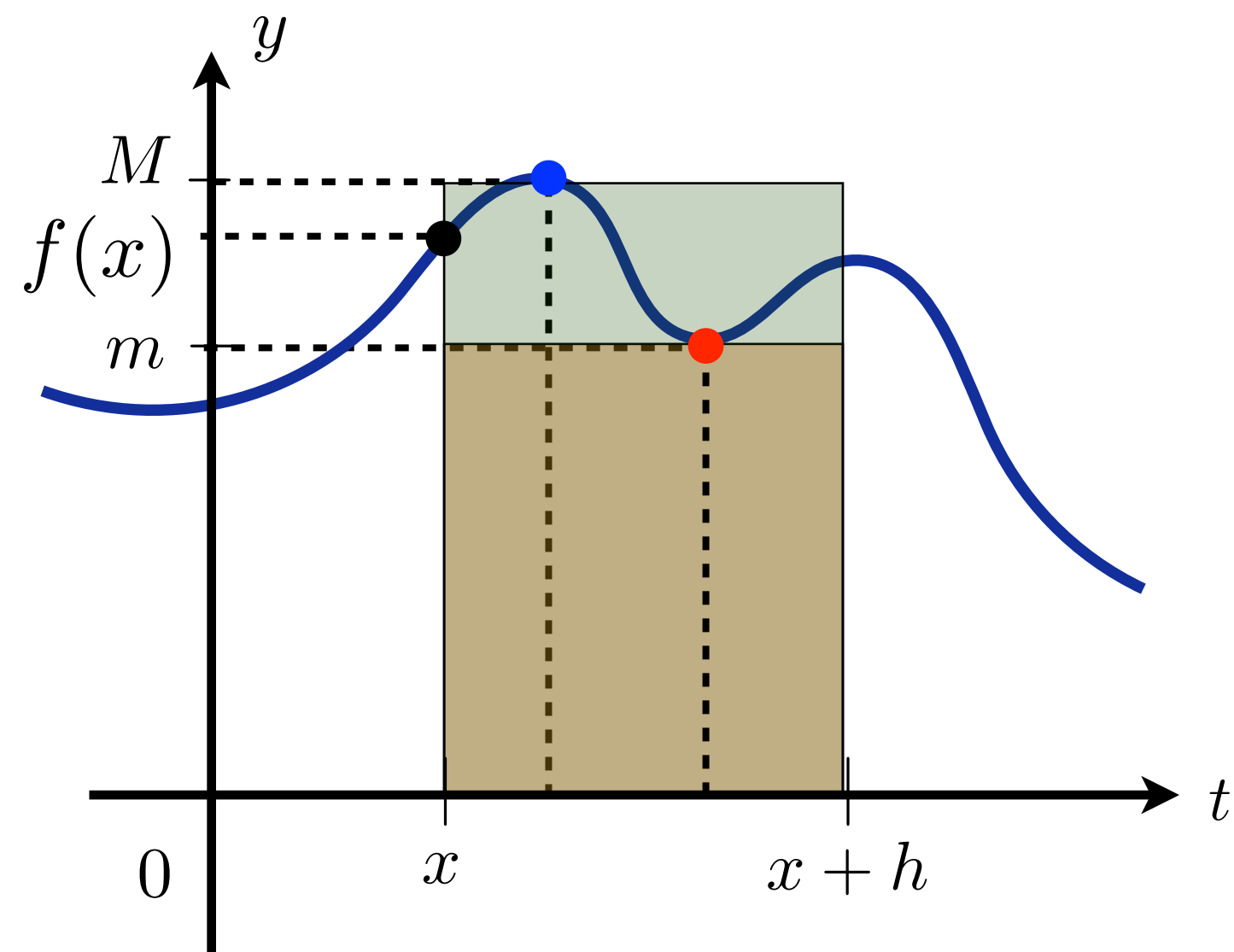
The Fundamental Theorem of Calculus

Theorem (The Fundamental Theorem of Calculus, Part I): If f is continuous on $[a, b]$, then $F(x) = \int_a^x f(t)dt$ is continuous on $[a, b]$ and differentiable on (a, b) and its derivative is $f(x)$:

$$F'(x) = \frac{d}{dx} \int_a^x f(t)dt = f(x)$$

Proof idea.

$$\begin{aligned} \text{Green box: } A_{\max} &= hM; & M &= \max_{t \in [x, x+h]} f(t) \\ \text{Brown box: } A_{\min} &= hm; & m &= \min_{t \in [x, x+h]} f(t) \end{aligned}$$



Definition of derivative:

$$\begin{aligned} \frac{d}{dx} \int_a^x f(t)dt &= \lim_{h \rightarrow 0} \frac{1}{h} \left(\overbrace{\int_a^{x+h} f(t)dt}^{F(x+h)} - \overbrace{\int_a^x f(t)dt}^{F(x)} \right) \\ &= \lim_{h \rightarrow 0} \frac{1}{h} \int_x^{x+h} f(t)dt \\ \text{Estimate integral: } m &= \frac{A_{\min}}{h} \leq \frac{1}{h} \int_x^{x+h} f(t)dt \leq \frac{A_{\max}}{h} = M \end{aligned}$$

Take limit as $h \rightarrow 0$:

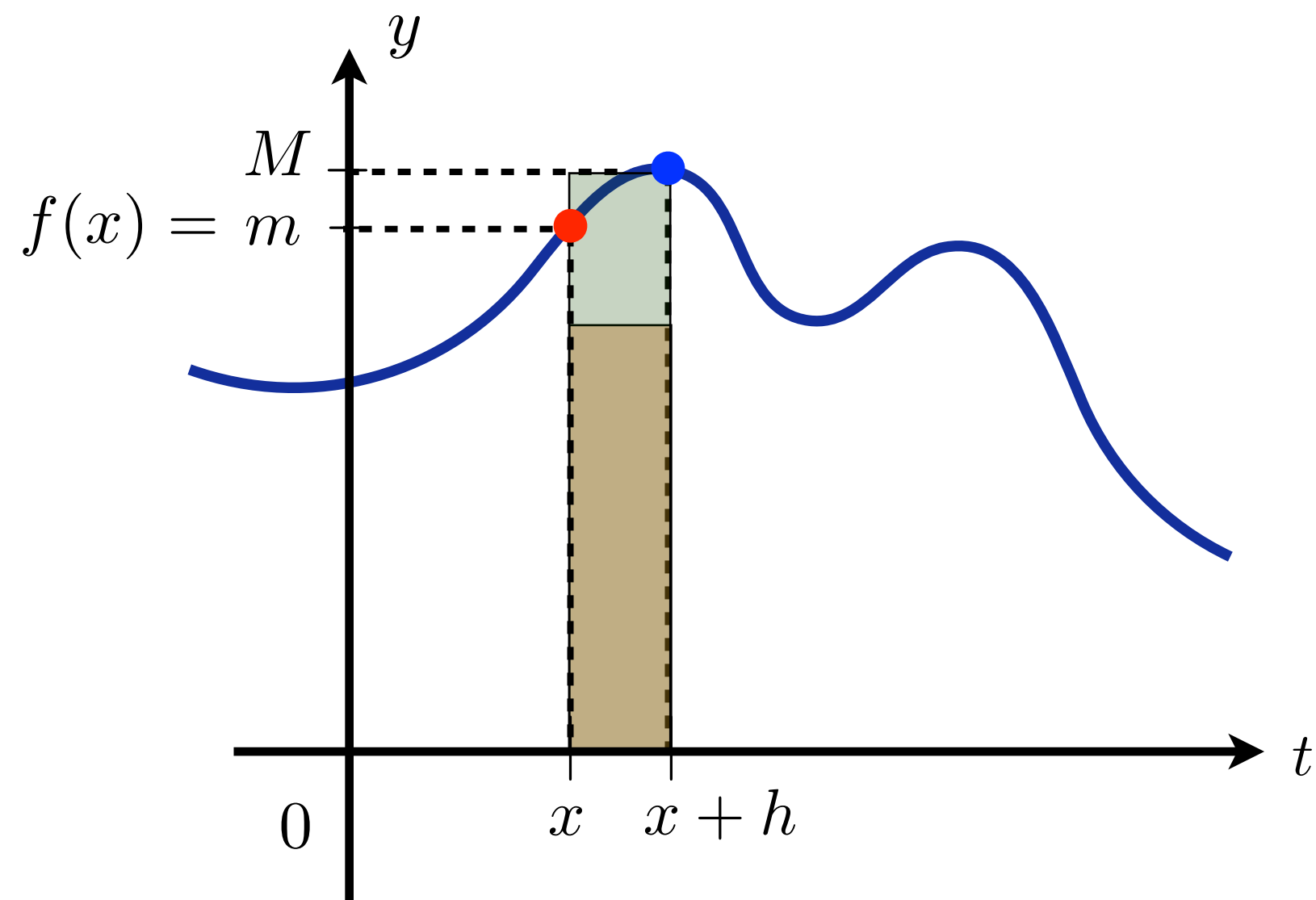
The Fundamental Theorem of Calculus

Theorem (The Fundamental Theorem of Calculus, Part I): If f is continuous on $[a, b]$, then $F(x) = \int_a^x f(t)dt$ is continuous on $[a, b]$ and differentiable on (a, b) and its derivative is $f(x)$:

$$F'(x) = \frac{d}{dx} \int_a^x f(t)dt = f(x)$$

Proof idea.

$\text{■ } A_{\max} = hM; \quad M = \max_{t \in [x, x+h]} f(t)$
 $\text{■ } A_{\min} = hm; \quad m = \min_{t \in [x, x+h]} f(t)$



Definition of derivative:

$$\begin{aligned} \frac{d}{dx} \int_a^x f(t)dt &= \lim_{h \rightarrow 0} \frac{1}{h} \left(\overbrace{\int_a^{x+h} f(t)dt}^{F(x+h)} - \overbrace{\int_a^x f(t)dt}^{F(x)} \right) \\ &= \lim_{h \rightarrow 0} \frac{1}{h} \int_x^{x+h} f(t)dt \end{aligned}$$

Estimate integral:

$$m = \frac{A_{\min}}{h} \leq \frac{1}{h} \int_x^{x+h} f(t)dt \leq \frac{A_{\max}}{h} = M$$

Take limit as $h \rightarrow 0$:

$$f(x) \leq \underbrace{\lim_{h \rightarrow 0} \frac{1}{h} \int_x^{x+h} f(t)dt}_{\frac{d}{dx} \int_a^x f(t)dt} \leq f(x)$$

The Fundamental Theorem of Calculus

Example: Use the Fundamental Theorem of Calculus to find $\frac{dy}{dx}$ in each case:

$$(a) \ y = \int_x^5 3t \sin t \, dt, \qquad (b) \ y = \int_{1+3x^2}^4 \frac{1}{2+e^t} \, dt$$

Solution.

(a) Since the lower limit depends on x , we rewrite the integral and apply the FTC:

$$\frac{d}{dx} \int_x^5 3t \sin t \, dt = -\frac{d}{dx} \int_5^x 3t \sin t \, dt = -3x \sin(x).$$

(b) Using

$$\frac{d}{dx} \int_{g(x)}^c f(t) \, dt = -\frac{d}{dx} \int_c^{g(x)} f(t) \, dt = -\frac{d}{dx} F(g(x)) = -f(g(x))g'(x),$$

with $f(t) = \frac{1}{2+e^t}$ and $g(x) = 1 + 3x^2$ ($g'(x) = 6x$), we get: $\frac{dy}{dx} = -\frac{6x}{2+e^{1+3x^2}}.$

Antiderivatives

Definition: A function F is an **antiderivative** of f on an interval I if

$$F'(x) = f(x)$$

for all x in I .

Observation: If F is an antiderivative of f on an interval I , then the most general antiderivative of f on I is

$$F(x) + C$$

where C is an arbitrary constant.

Example: Find an antiderivative of $f(x) = 3x^2$ that satisfies $F(1) = -1$.

Solution. An antiderivative of $f(x) = 3x^2$ is a function $F(x)$ such that $F'(x) = 3x^2$. We know that the derivative of x^3 is $3x^2$, so

$$F(x) = x^3 + C.$$

Using $F(1) = -1$ gives $1^3 + C = -1 \implies C = -2$. Therefore, the required antiderivative is

$$F(x) = x^3 - 2.$$

The Fundamental Theorem of Calculus

Theorem (The Fundamental Theorem of Calculus, Part II): If f is continuous at every point in $[a, b]$ and F is any antiderivative of f on $[a, b]$, then

$$\int_a^b f(x)dx = F(b) - F(a)$$

Proof idea. Define $G(x) := \int_a^x f(t)dt$, $x \in [a, b]$. By the FTC-I, $G'(x) = f(x)$.

Let F be any another antiderivatives of f . Then

$$F(x) = G(x) + C$$

for some constant C .

Since $G(a) = 0$ and $G(b) = \int_a^b f(x)dx$, we obtain

$$\int_a^b f(x)dx = G(b) - G(a) = (G(b) + C) - (G(a) + C) = F(b) - F(a).$$

Indefinite Integral

The **Fundamental Theorem of Calculus** tells us that definite integrals can be computed using antiderivatives, which motivates the definition of the **indefinite integral**.

Definition: The indefinite integral of a function f with respect to x is the set of all antiderivatives of f :

$$\int f(x) dx.$$

General Form:

$$\int f(x) dx = F(x) + C, \quad \text{where } F'(x) = f(x) \text{ and } C \text{ is arbitrary.}$$

Key Distinction:

- Definite integral $\int_a^b f(x) dx \rightarrow$ a **number**.
- Indefinite integral $\int f(x) dx \rightarrow$ a **family of functions**.

Change of Variables/Substitution

Theorem (The Substitution Rule): If $u = g(x)$ is a differentiable function whose range is an interval I , and f is continuous on I , then

$$\int f(g(x))g'(x)dx = \int f(u)du$$

Proof idea. If $F'(x) = f(x)$, then

$$\frac{d}{dx}F(g(x)) = F'(g(x)) \cdot g'(x) = f(g(x))g'(x).$$

If we make the substitution $u = g(x)$, then

$$\begin{aligned}\int f(g(x))g'(x) dx &= \int \frac{d}{dx}F(g(x)) dx \\ &= F(g(x)) + C && \text{(Fundamental Theorem)} \\ &= F(u) + C && (u = g(x)) \\ &= \int F'(u) du && \text{(Fundamental Theorem)} \\ &= \int f(u) du.\end{aligned}$$

Change of Variables/Substitution

The **Substitution Rule** provides a method to evaluate integrals of the form

$$\int f(g(x))g'(x) dx,$$

when f and g are continuous functions.

Procedure:

1. Make the substitution $u = g(x)$, so that

$$du = \frac{du}{dx} dx = g'(x) dx,$$

and the integral becomes

$$\int f(u) du.$$

2. Integrate with respect to u .
3. Replace u by $g(x)$ in the result.

Change of Variables/Substitution

Example. Find $\int \cos(7\theta + 3)d\theta$.

Solution. Let $u = 7\theta + 3$. Then $du = 7d\theta \implies d\theta = \frac{1}{7} du$. Replacing in the integral:

$$\int \cos(7\theta + 3) d\theta = \frac{1}{7} \int \cos(u) du = \frac{1}{7} \sin(u) + C = \int \cos(7\theta + 3) d\theta = \frac{1}{7} \sin(7\theta + 3) + C.$$

Example. Find $\int x^2 e^{x^3} dx$.

Solution. Let $u = x^3$. Then $du = 3x^2 dx \implies x^2 dx = \frac{1}{3} du$. Replacing in the integral:

$$\int x^2 e^{x^3} dx = \frac{1}{3} \int e^u du = \frac{1}{3} e^u + C = \frac{1}{3} e^{x^3} + C.$$

Example. Find $\int x\sqrt{2x+1}dx$.

Solution. Let $u = 2x + 1$. Then $du = 2dx \implies dx = \frac{1}{2} du$, and $x = \frac{u-1}{2}$. Replacing in the integral:

$$\begin{aligned} \int x\sqrt{2x+1} dx &= \int \frac{u-1}{2} \cdot \sqrt{u} \cdot \frac{1}{2} du = \frac{1}{4} \int (u-1)u^{1/2} du = \frac{1}{4} \int (u^{3/2} - u^{1/2}) du \\ &= \frac{1}{4} \left(\frac{2}{5} u^{5/2} - \frac{2}{3} u^{3/2} \right) + C = \frac{1}{10} u^{5/2} - \frac{1}{6} u^{3/2} + C = \frac{1}{10} (2x+1)^{5/2} - \frac{1}{6} (2x+1)^{3/2} + C. \end{aligned}$$

Change of Variables/Substitution

Theorem (Substitution in Definite Integrals): If g' is continuous on the interval $[a, b]$ and f is continuous on the range of $g(x) = u$, then

$$\int_a^b f(g(x)) \cdot g'(x) dx = \int_{g(a)}^{g(b)} f(u) du$$

Proof.

$$\int_a^b f(g(x)) g'(x) dx = F(g(x)) \Big|_{x=a}^{x=b} = F(g(b)) - F(g(a)) = F(u) \Big|_{u=g(a)}^{u=g(b)} = \int_{g(a)}^{g(b)} f(u) du.$$

Example. Evaluate $\int_{-1}^1 3x^2 \sqrt{x^3 + 1} dx$

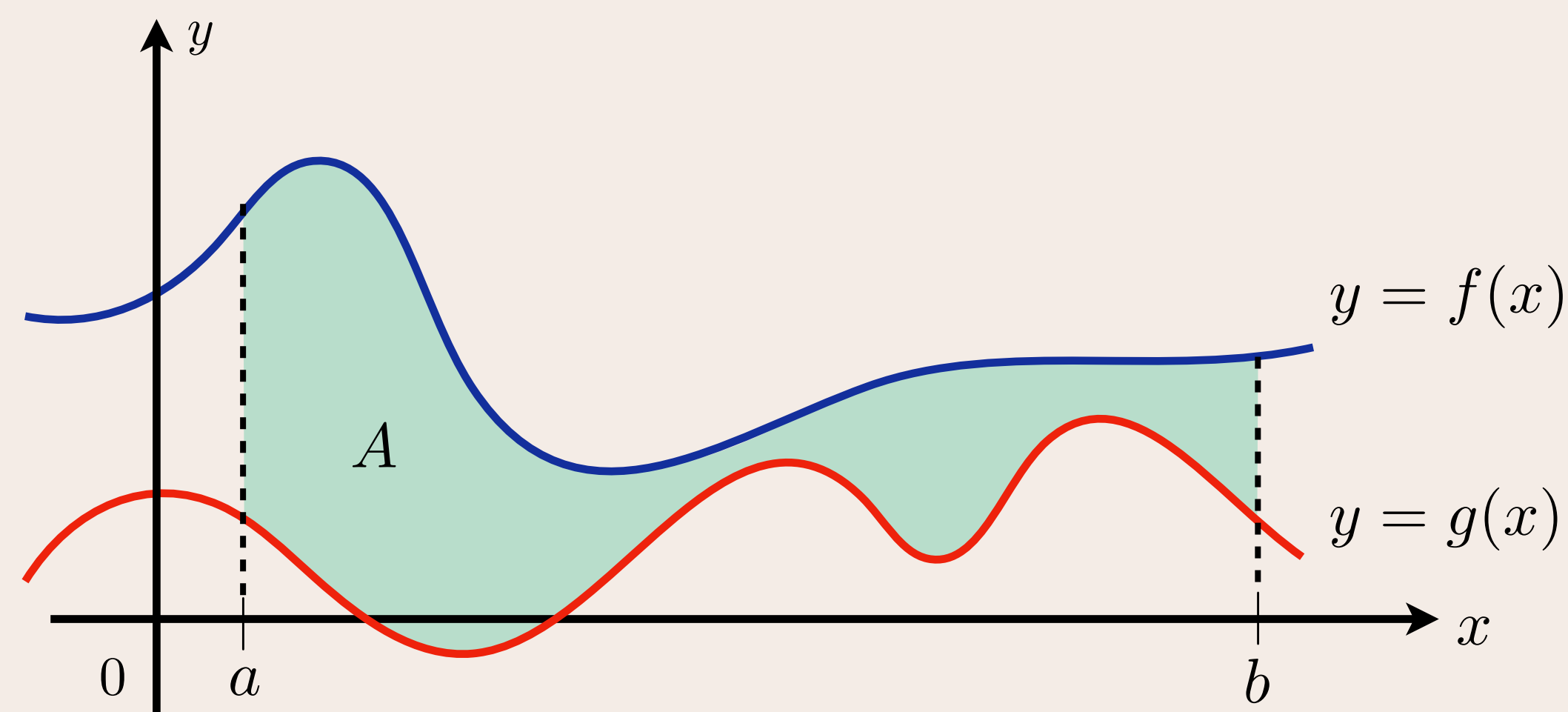
Solution. Let $u = x^3 + 1 \Rightarrow du = 3x^2 dx$. When $x = -1$, $u = 0$; when $x = 1$, $u = 2$. Thus:

$$\int_{-1}^1 3x^2 \sqrt{x^3 + 1} dx = \int_0^2 \sqrt{u} du = \frac{2}{3} u^{3/2} \Big|_0^2 = \frac{2}{3} (2^{3/2}) = \frac{4}{3} \sqrt{2}.$$

Areas Between Curves

Definition: If f and g are continuous with $f(x) \geq g(x)$ throughout $[a, b]$, then the area of the region between the curves $y = f(x)$ and $y = g(x)$ from a to b is the integral of $(f - g)$ from a to b :

$$A = \int_a^b [f(x) - g(x)] dx$$

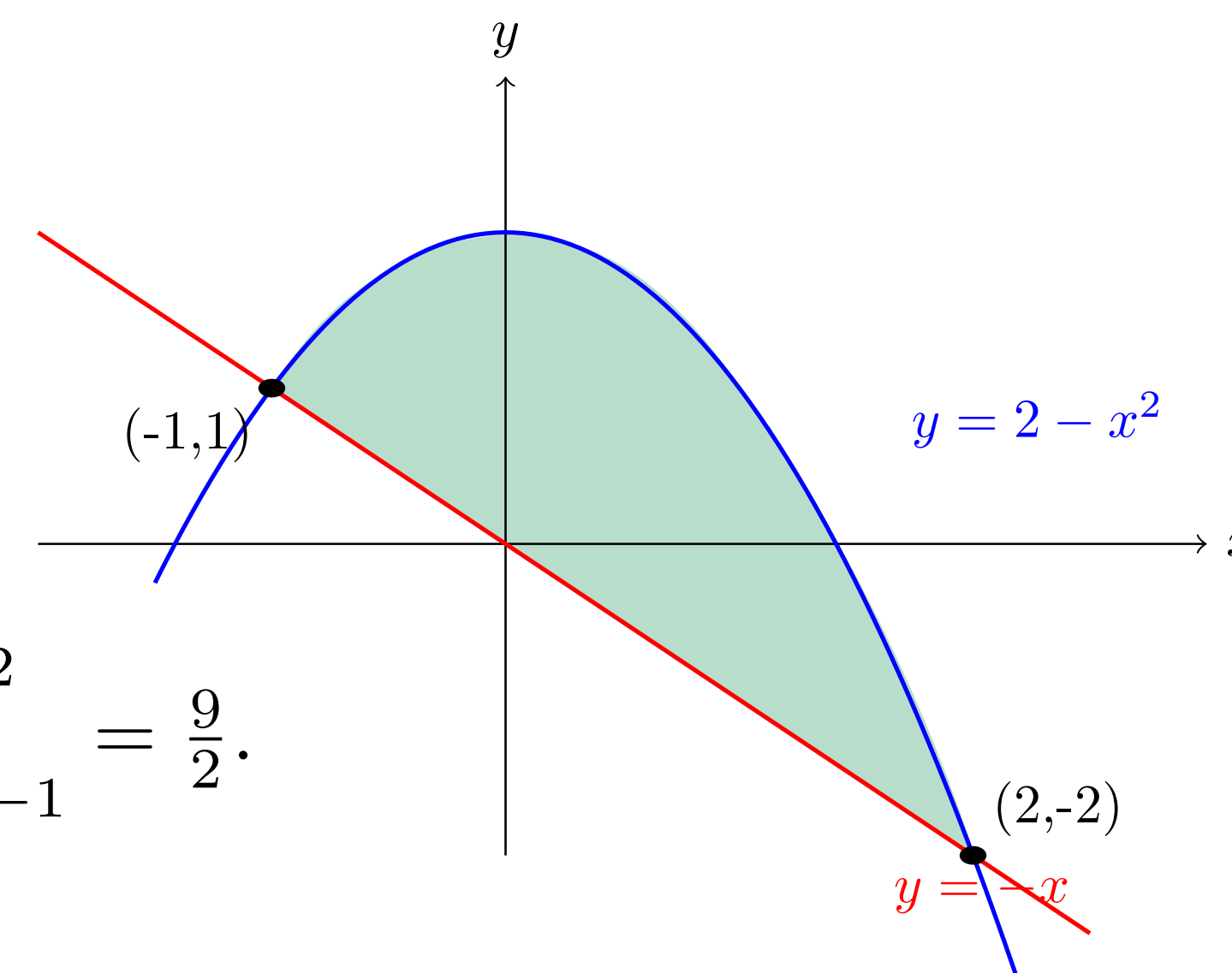


Example: Find the area of the region enclosed by the parabola $y = 2 - x^2$ and the line $y = -x$.

Solution.

Find intersections: $2 - x^2 = -x \Rightarrow x^2 - x - 2 = 0 \Rightarrow x = -1, 2$.

$$\text{Area: } A = \int_{-1}^2 [(2 - x^2) - (-x)] dx = \int_{-1}^2 (2 + x - x^2) dx = \left[2x + \frac{x^2}{2} - \frac{x^3}{3} \right]_{-1}^2 = \frac{9}{2}.$$



Integration by Parts

Product Rule: If f and g are differentiable,

$$\frac{d}{dx}[f(x)g(x)] = f'(x)g(x) + f(x)g'(x).$$

In terms of indefinite integrals:

$$\int \frac{d}{dx}[f(x)g(x)] dx = \int f'(x)g(x) dx + \int f(x)g'(x) dx.$$

Rearranging gives the **Integration by Parts** formula:

$$\int f(x)g'(x) dx = f(x)g(x) - \int f'(x)g(x) dx.$$

Example: Find $\int x \cos x dx$. **Solution.** Let $f(x) = x$, $f'(x) = 1$; $g'(x) = \cos x$, $g(x) = \sin x$. Then

$$\boxed{\int x \cos x dx} = fg - \int f'g dx = x \sin x - \boxed{\int \sin x dx} = x \sin x + \cos x + C$$

Key idea: Transform a difficult integral into a simpler one by transferring the derivative from one function to the other.

Integration by Parts

Example: Find $\int \ln x dx$.

Solution. Let $f(x) = \ln x$, $f'(x) = \frac{1}{x}$, $g'(x) = 1$, $g(x) = x$. Then

$$\int \ln x dx = x \ln x - \int x \cdot \frac{1}{x} dx = x \ln x - \int 1 dx = x \ln x - x + C$$

Example: Find $\int x^2 e^x dx$.

Solution. Let $f(x) = x^2$, $f'(x) = 2x$, $g'(x) = e^x$, $g(x) = e^x$. Then, applying the integration by parts formula we get

$$\int x^2 e^x dx = x^2 e^x - \int 2x e^x dx$$

Applying integration by parts again on $\int 2x e^x dx$:

$$\int 2x e^x dx = 2(x e^x - \int e^x dx) = 2(x e^x - e^x)$$

Combining the results we arrive at:

$$\int x^2 e^x dx = x^2 e^x - 2(x e^x - e^x) = x^2 e^x - 2x e^x + 2e^x + C$$

Trigonometric Integrals

Strategy depends on whether the power is even or odd:

- Odd power of sin: Save one $\sin x$, use $\sin^2 x = 1 - \cos^2 x$, and substitute $u = \cos x$.

Example:
$$\int \sin^3 x \, dx = \int (1 - \cos^2 x) \sin x \, dx = - \int (1 - u^2) du \quad (u = \cos x, \, du = -\sin x \, dx)$$

$$= - \left(u - \frac{u^3}{3} \right) + C = -\cos x + \frac{\cos^3 x}{3} + C.$$

- Odd power of cos: Save one $\cos x$, use $\cos^2 x = 1 - \sin^2 x$, and substitute $u = \sin x$.
- Even powers: Use half-angle identities

$$\sin^2 x = \frac{1}{2}(1 - \cos 2x),$$

$$\cos^2 x = \frac{1}{2}(1 + \cos 2x).$$

Example:
$$\int \sin^2 x \, dx = \frac{1}{2} \int (1 - \cos 2x) \, dx = \frac{x}{2} - \frac{\sin 2x}{4} + C.$$

Example:
$$\int \cos^2 x \, dx = \frac{1}{2} \int (1 + \cos 2x) \, dx = \frac{x}{2} + \frac{\sin 2x}{4} + C.$$

Key idea: Reduce the power until only basic trigonometric integrals remain.

Improper Integrals

Although some regions extend infinitely or involve unbounded functions, we can still assign them a precise finite “area” using improper integrals.

Example. Consider the curve $y = 1/x^2$ for $x \geq 1$. Since the region extends infinitely, we define the area as an **improper integral**:

$$\int_1^{\infty} \frac{1}{x^2} dx := \lim_{b \rightarrow \infty} \int_1^b \frac{1}{x^2} dx.$$

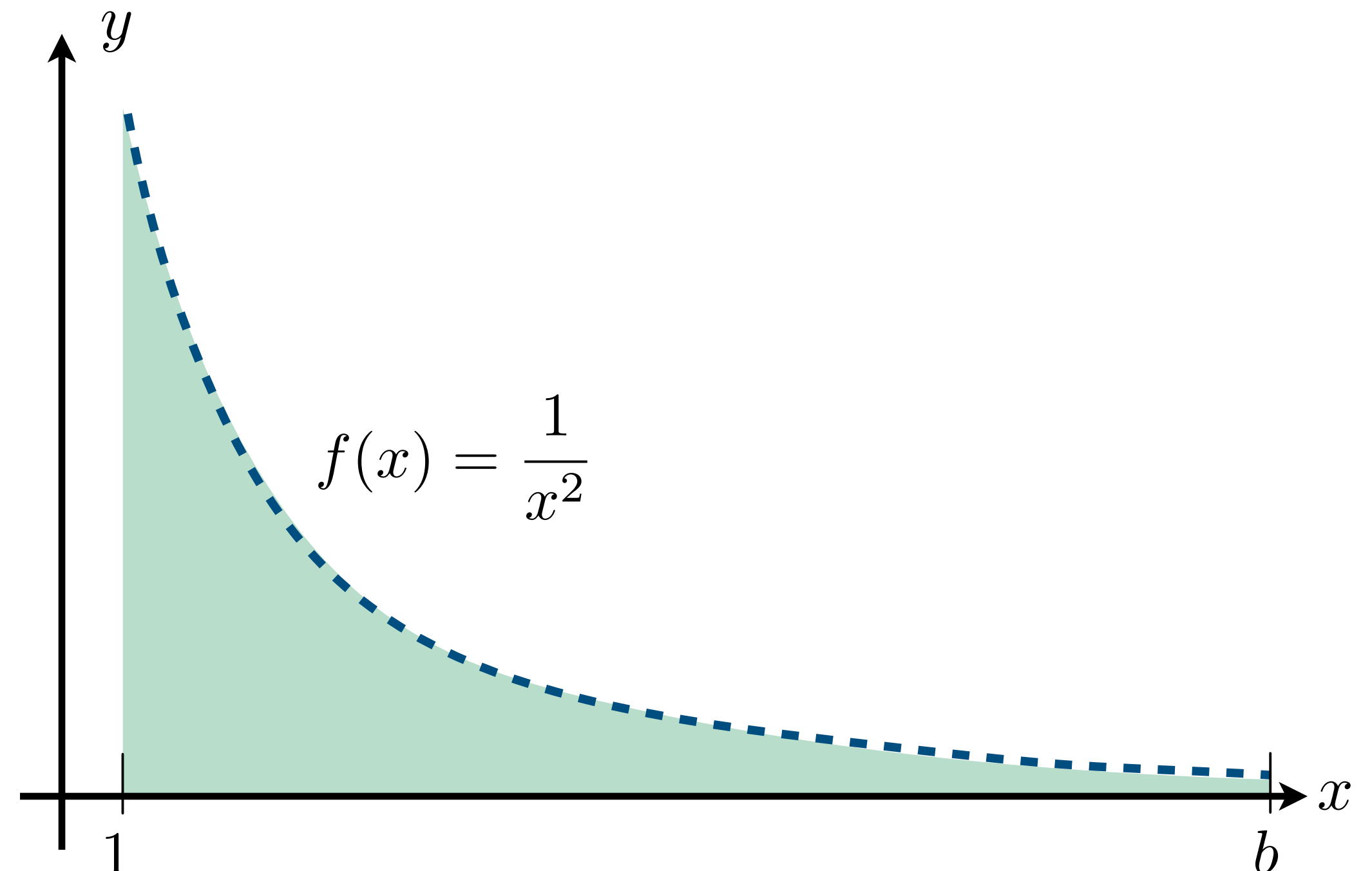
For finite b :

$$\int_1^b \frac{1}{x^2} dx = \left[-\frac{1}{x} \right]_1^b = 1 - \frac{1}{b}.$$

Taking the limit:

$$\int_1^{\infty} \frac{1}{x^2} dx = \lim_{b \rightarrow \infty} \left(1 - \frac{1}{b} \right) = 1.$$

Even though the region is infinite, its total area is finite.



Improper Integrals

Definition: Integrals with infinite limits of integration are called **improper integrals of Type I**.

1. If $f(x)$ is continuous on $[a, \infty)$, then

$$\int_a^{\infty} f(x) dx = \lim_{b \rightarrow \infty} \int_a^b f(x) dx.$$

2. If $f(x)$ is continuous on $(-\infty, b]$, then

$$\int_{-\infty}^b f(x) dx = \lim_{a \rightarrow -\infty} \int_a^b f(x) dx.$$

3. If $f(x)$ is continuous on $(-\infty, \infty)$, then

$$\int_{-\infty}^{\infty} f(x) dx = \int_{-\infty}^c f(x) dx + \int_c^{\infty} f(x) dx,$$

where c is any real number.

In each case, if the limit exists and is finite, we say the improper integral **converges** and its value is the limit. If the limit does not exist, the improper integral **diverges**.

Improper Integrals

Example: For what values of p does the integral

$$\int_1^{\infty} \frac{dx}{x^p}$$

converge? When it does converge, what is its value?

Improper Integrals

Solution. Consider $p \neq 1$. Evaluate the improper integral as a limit:

$$\int_1^{\infty} x^{-p} dx = \lim_{b \rightarrow \infty} \int_1^b x^{-p} dx = \lim_{b \rightarrow \infty} \left[\frac{x^{1-p}}{1-p} \right]_1^b = \lim_{b \rightarrow \infty} \frac{b^{1-p} - 1}{1-p}.$$

- If $p > 1$, then $1 - p < 0$, so $b^{1-p} \rightarrow 0$ as $b \rightarrow \infty$. Hence

$$\int_1^{\infty} x^{-p} dx = \frac{0 - 1}{1 - p} = \frac{1}{p - 1}.$$

- If $p < 1$, then $1 - p > 0$, so $b^{1-p} \rightarrow \infty$ and the integral diverges.

For $p = 1$ we have the logarithmic case:

$$\int_1^{\infty} \frac{dx}{x} = \lim_{b \rightarrow \infty} \ln b = \infty,$$

so the integral diverges.

Conclusion. The integral converges if and only if $p > 1$, and when it converges

$$\int_1^{\infty} \frac{dx}{x^p} = \frac{1}{p - 1} \quad (p > 1).$$

Improper Integrals

Definition: Integrals of functions that become infinite at a point within the interval of integration are called **improper integrals of Type II**.

1. If $f(x)$ is continuous on $(a, b]$ and has a discontinuity at a , then

$$\int_a^b f(x) dx = \lim_{c \rightarrow a^+} \int_c^b f(x) dx.$$

2. If $f(x)$ is continuous on $[a, b)$ and has a discontinuity at b , then

$$\int_a^b f(x) dx = \lim_{c \rightarrow b^-} \int_a^c f(x) dx.$$

3. If $f(x)$ is discontinuous at c , where $a < c < b$, and continuous on $[a, c) \cup (c, b]$, then

$$\int_a^b f(x) dx = \int_a^c f(x) dx + \int_c^b f(x) dx.$$

In each case, if the limit is finite, the improper integral is said to **converge**, and its value is the limit. If the limit does not exist, the improper integral **diverges**.

Improper Integrals

Example: Evaluate $\int_0^3 \frac{dx}{(x-1)^{2/3}}$.

Solution. The integrand is not defined at $x = 1$. We split the integral and take limits:

$$\int_0^3 \frac{dx}{(x-1)^{2/3}} = \lim_{b \rightarrow 1^-} \int_0^b \frac{dx}{(1-x)^{2/3}} + \lim_{a \rightarrow 1^+} \int_a^3 \frac{dx}{(x-1)^{2/3}}.$$

Step 1: Left-hand limit. Let $u = 1 - x$, $du = -dx$: When $x = 0$, $u = 1$; when $x = b$, $u = 1 - b$. Then

$$\lim_{b \rightarrow 1^-} \int_0^b \frac{dx}{(1-x)^{2/3}} = \lim_{b \rightarrow 1^-} \int_1^{1-b} u^{-2/3}(-du) = \lim_{b \rightarrow 1^-} \int_{1-b}^1 u^{-2/3} du = \lim_{b \rightarrow 1^-} 3u^{1/3} \Big|_{1-b}^1 = 3 - 0 = 3.$$

Step 2: Right-hand limit. Let $v = x - 1$, $dv = dx$:

$$\lim_{a \rightarrow 1^+} \int_a^3 \frac{dx}{(x-1)^{2/3}} = \lim_{a \rightarrow 1^+} \int_{a-1}^2 v^{-2/3} dv = \lim_{a \rightarrow 1^+} 3v^{1/3} \Big|_{a-1}^2 = 3 \cdot 2^{1/3}.$$

Conclusion: The integral converges and

$$\int_0^3 \frac{dx}{(x-1)^{2/3}} = 3 + 3 \cdot 2^{1/3} = 3(1 + 2^{1/3}).$$